



COMSATS University Islamabad

Department of Computer Science

Course Information

Course Code: **CSC354**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Machine Learning**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

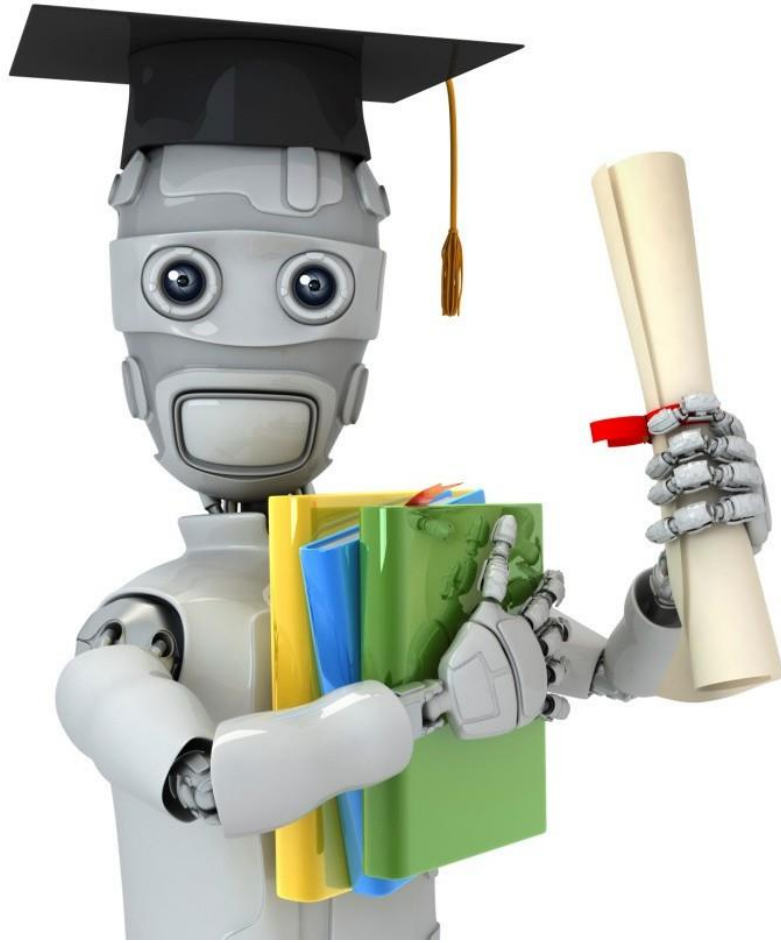
Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O'Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.



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- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**



Machine Learning

Introduction

Unsupervised Learning Model (PCA)

Principal Component Analysis

- **PCA stands for Principal Component Analysis.**
- **PCA is an unsupervised learning technique** because it works without class labels.
- **PCA is mainly used for dimensionality reduction**, reducing the number of features in a dataset.
- **Dimensionality** means the **number of features (variables)** in a dataset.

For example:

Student	Math Marks	Physics Marks	Chemistry Marks	English Marks	Attendance
A	85	80	82	78	90

Principal Component Analysis

- This dataset has **5 features (dimensions)**:
- Math Marks
- Physics Marks
- Chemistry Marks
- English Marks
- Attendance
- So, the dimensionality is **5**.

Principal Component Analysis

What PCA Does

- Suppose Math, Physics, and Chemistry marks are highly related (students who score high in one often score high in the others).
- Instead of keeping all three features separately, PCA can combine them into a single new feature called a **Principal Component**.

Student	PC1 (Science Performance)	English Marks	Attendance
A	82.3	78	90

Principal Component Analysis

- Now the dataset has only **3 features instead of 5**.
- This process is called **dimensionality reduction**.

Student	PC1 (Science Performance)	English Marks	Attendance
A	82.3	78	90

Principal Component Analysis

- **PCA can reduce overfitting** by removing less important features and noise.
- **It helps reduce computational cost** by decreasing the number of input features.
- **PCA is commonly used in image processing, pattern recognition, and cybersecurity. It works best when features have linear relationships.**

Principal Component Analysis

- Removes less important information.
- Keeps the most useful information.
- Reduces the number of features.
- It transforms original features into new features called Principal Components (PCs).

Principal Component Analysis

- PCA visualization helps to convert 3D data into a 2D format or even 2D data to a one-dimensional representation.
- The main goal of principal component analysis is visualizing data in a lower dimensional space.

Principal Component Analysis

What Type of Dataset Does PCA Work Well On?

1. Highly Correlated Features

PCA works very well when several features contain similar information.

Example:

Height

Weight

Body Mass Index (BMI)

These features are related, so PCA can combine them into fewer principal components.

Principal Component Analysis

What Type of Dataset Does PCA Work Well On?

2. Numerical Data

PCA is designed for **continuous numerical variables**.

Good Features:

Age

Income

Temperature

Sensor readings

Not Ideal:

Gender (Male/Female)

Country names

Colors

Point	X_1	X_2
A	1	2
B	3	4
C	5	6

Principal
Component
Analysis

Principal Component Analysis

- Apply **PCA** to reduce this dataset from 2D to 1D (PC1 only).
- **STEP 1: Write the Dataset Matrix**

- $X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

Principal Component Analysis

STEP 2: Compute Mean of Each Feature

- **For X_1 :**

- $\mu_1 = \frac{1+3+5}{3} = \frac{9}{3} = 3$

- **For X_2 :**

- $\mu_2 = \frac{2+4+6}{3} = \frac{12}{3} = 4$

Principal Component Analysis

- **STEP 3: Mean Centering (Standardization Step 1)**
- Subtract mean from each value:
- **Point A:**
 - $X_1 = 1 - 3 = -2$
 - $X_2 = 2 - 4 = -2$
- **Point B:**
 - $X_1 = 3 - 3 = 0$
 - $X_2 = 4 - 4 = 0$
- **Point C:**
 - $X_1 = 5 - 3 = 2$
 - $X_2 = 6 - 4 = 2$

Principal Component Analysis

- **Mean-centered matrix:**

- $X_{centered} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$

Principal Component Analysis

- **STEP 4: Compute Covariance Matrix**

- Formula:

- $$\mathbf{C} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$$

- Here:

- $n = 3 \rightarrow n - 1 = 2$

- **Step 4.1: Compute $\mathbf{X}^T$**

- $$\mathbf{X}^T = \begin{bmatrix} -2 & 0 & 2 \\ -2 & 0 & 2 \end{bmatrix}$$

Principal Component Analysis

- **Step 4.2: Multiply $X^T X$**
- **First element:**
- $(-2)^2 + 0^2 + 2^2 = 4 + 0 + 4 = 8$ So $(1,1) = 8$
- $(-2)(-2) + (0)(0) + (2)(2) = 8$ So $(1,2) = 8$
- So $(2,1) = 8$
- So $(2,2) = 8$

- So:
- $X^T X = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$
- **Step 4.3: Divide by (n-1 = 2)**
- $C = \frac{1}{2} \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

Principal Component Analysis

- **STEP 5: Find Eigenvalues**

- Solve:

- $|C - \lambda I| = 0$

- $\begin{vmatrix} 4 - \lambda & 4 \\ 4 & 4 - \lambda \end{vmatrix} = 0$

Identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply by λ :

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Principal Component Analysis

- **Determinant:**

- $(4 - \lambda)(4 - \lambda) - 16 = 0$

- $(4 - \lambda)^2 - 16 = 0$

- **Expand:**

- $(4 - \lambda)^2 = 16$

- Take square root:

- $4 - \lambda = \pm 4$

Principal Component Analysis

- **Case 1:**
 - $4 - \lambda = 4 \Rightarrow \lambda = 0$
- **Case 2:**
 - $4 - \lambda = -4 \Rightarrow \lambda = 8$
- **Eigenvalues:**
 - $\lambda_1 = 8, \lambda_2 = 0$

Principal Component Analysis

STEP 6: Find Eigenvectors

- **For $\lambda = 8$:**
- $(C - 8I)v = 0$
- **Substitute $\lambda = 8$**
- $C - 8I = \begin{bmatrix} 4 - 8 & 4 \\ 4 & 4 - 8 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 4 & -4 \end{bmatrix}$
- Equation:
- From first row:
- $-4x + 4y = 0$
- From second row:
- $4x - 4y = 0$
- Divide by 4:
- $-x + y = 0$
- $y = x$

Principal Component Analysis

- **Eigenvector:**
- $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Any vector where both values are equal is an eigenvector direction.

Principal Component Analysis

- **Normalize Eigenvector:**

- Find magnitude (length)

- $\|v\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

- To normalize:

- $v_{normalized} = \frac{v}{\|v\|}$

- So:

- $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

- $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\|v\| = \sqrt{x^2 + y^2}$$

Principal Component Analysis

- **STEP 7: Select Principal Component (PC1)**
- Since:
- $\lambda_1 = 8$ (largest eigenvalue)
- So:
- $PC1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Principal Component Analysis

- **STEP 8: Transform Data into 1D (Projection)**

- Formula:

- $Y = X_{centered} \cdot v_1$

- **For A:**

- $(-, 2 - 2) \cdot \frac{1}{\sqrt{2}} (1, 1)$

- $= \frac{-2-2}{\sqrt{2}} = \frac{-4}{\sqrt{2}} = -2.828$

Principal Component Analysis

- **For B:**

- $(0, 0) \Rightarrow 0$

- **For C:**

- $(2, 2) \cdot \frac{1}{\sqrt{2}} (1, 1)$

- $= \frac{4}{\sqrt{2}} = 2.828$

Principal Component Analysis

Point	PC1
A	-2.828
B	0
C	2.828

Principal Component Analysis

- Principal Component Analysis is an unsupervised learning algorithm that is used for the dimensionality reduction in machine learning. It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation. These new transformed features are called the Principal Components. It is one of the popular tools that is used for exploratory data analysis and predictive modeling. It is a technique to draw strong patterns from the given dataset by reducing the variances.
- PCA generally tries to find the lower-dimensional surface to project the high-dimensional data.
- PCA works by considering the variance of each attribute because the high attribute shows a good split between the classes, and hence it reduces the dimensionality. Some real-world applications of PCA are **image processing, movie recommendation systems, and optimizing the power allocation in various communication channels**. It is a feature extraction technique, so it contains the important variables and drops the least important variable.

Principal Component Analysis

- The PCA algorithm is based on some mathematical concepts such as:
 - Variance and Covariance
 - Eigenvalues and Eigen factors
- Some common terms used in the PCA algorithm:
 - **Dimensionality:** It is the number of features or variables present in the given dataset. More easily, it is the number of columns present in the dataset.
 - **Correlation:** It signifies how strongly two variables are related to each other. Such as if one changes, the other variable also gets changed. The correlation value ranges from -1 to +1. Here, -1 occurs if variables are inversely proportional to each other, and +1 indicates that variables are directly proportional to each other.

Principal Component Analysis

- Orthogonal:** It defines that variables are not correlated to each other, and hence the correlation between the pair of variables is zero.
- Eigenvectors:** If there is a square matrix M and a non-zero vector v is given. Then v will be the eigenvector if $Mv = \lambda v$ is the scalar multiple of v .
- Covariance Matrix:** A matrix containing the covariance between the pair of variables is called the Covariance Matrix.

Steps Principal Component Analysis

1. Getting the dataset

Firstly, we need to take the input dataset and divide it into two subparts X and Y, where X is the training set, and Y is the validation set.

2. Representing data into a structure

Now we will represent our dataset into a structure. Such as we will represent the two-dimensional matrix of independent variable X. Here each row corresponds to the data items, and the column corresponds to the Features. The number of columns is the dimensions of the dataset.

3. Standardizing the data

In this step, we will standardize our dataset. Such as in a particular column, the features with high variance are more important compared to the features with lower variance.

If the importance of features is independent of the variance of the feature, then we will divide each data item into a column with the standard deviation of the column. Here we will name the matrix as Z.

Steps Principal Component Analysis

4. Calculating the Covariance of Z

To calculate the covariance of Z, we will take the matrix Z, and will transpose it. After transpose, we will multiply it by Z. The output matrix will be the Covariance matrix of Z.

5. Calculating the Eigen Values and Eigen Vectors

Now we need to calculate the eigenvalues and eigenvectors for the resultant covariance matrix Z. Eigenvectors or the covariance matrix are the directions of the axes with high information. The coefficients of these eigenvectors are defined as the eigenvalues.

6. Sorting the Eigen Vectors

In this step, we will take all the eigenvalues and sort them in decreasing order, which means from largest to smallest. And simultaneously sort the eigenvectors accordingly in matrix P of eigenvalues. The resultant matrix will be named as P^* .

Steps Principal Component Analysis

7. Calculating the new features Or Principal Components

Here we will calculate the new features. To do this, we will multiply the P^* matrix to the Z . In the resultant matrix Z^* , each observation is the linear combination of original features. Each column of the Z^* matrix is independent of each other.

8. Remove less or unimportant features from the new dataset.

The new feature set has occurred, so we will decide here what to keep and what to remove. It means, we will only keep the relevant or important features in the new dataset, and unimportant features will be removed.

Principal Component Analysis Solved Example 1

- Given the data in Table, reduce the dimension from 2 to 1 using the Principal Component Analysis (PCA) algorithm.

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

Principal Component Analysis Solved Example 1

Step 1: Calculate Mean

$$\bar{X}_1 = \frac{1}{4}(4 + 8 + 13 + 7) = 8,$$

$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = 8.5.$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

Principal Component Analysis Solved Example 1

Step 2: Calculation of the covariance matrix.

$$S = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

$$\begin{aligned} \text{Cov}(X_1, X_1) &= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{1k} - \bar{X}_1) \\ &= \frac{1}{3} ((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2) \\ &= 14 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_1, X_2) &= \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2) \\ &= \frac{1}{3} ((4-8)(11-8.5) + (8-8)(4-8.5) \\ &\quad + (13-8)(5-8.5) + (7-8)(14-8.5)) \end{aligned}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

$$N = 4$$

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

Principal Component Analysis Solved Example 1

$$S = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$

$$\begin{aligned} \text{Cov}(X_2, X_1) &= \text{Cov}(X_1, X_2) \\ &= -11 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_2, X_2) &= \frac{1}{N-1} \sum_{k=1}^4 (X_{2k} - \bar{X}_2)(X_{2k} - \bar{X}_2) \\ &= \frac{1}{3} ((11 - 8.5)^2 + (4 - 8.5)^2 + (5 - 8.5)^2 + (14 - 8.5)^2) \\ &= 23 \end{aligned}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

Principal Component Analysis Solved Example 1

$$S = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$$
$$= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

Principal Component Analysis Solved Example 1

Step 3: Eigenvalues of the covariance matrix

The characteristic equation of the covariance matrix is,

Determinant

$$0 = \det(S - \lambda I)$$

$$= \begin{vmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{vmatrix}$$

$$= (14 - \lambda)(23 - \lambda) - (-11) \times (-11)$$

$$= \lambda^2 - 37\lambda + 201$$

$\Rightarrow$

$$0 = \lambda^2 - 37\lambda + 201$$

I is the identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

mean

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

Cov $\rightarrow$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Principal Component Analysis Solved Example 1

$$\lambda^2 - 37\lambda + 201 = 0 \rightarrow \text{Quadratic equation}$$

$$\Rightarrow a = 1, b = -37, c = 201$$

$$\Rightarrow b^2 - 4ac = (-37)^2 - 4(1)(201)$$

$$\Rightarrow 1369 - 804$$

$$\Rightarrow 565$$

$$\text{Assume, } X = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\text{So } \lambda = -\frac{-37}{2} \pm \frac{\sqrt{565}}{2}$$

$$\Rightarrow \lambda = \frac{37}{2} \pm \frac{\sqrt{565}}{2}$$

$$\Rightarrow \lambda = \frac{1}{2}(37 \pm \sqrt{565})$$

$$\Rightarrow \lambda_1 = 30.3849, \lambda_2 = 6.6151$$

Roots of quadratic equation

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

mean

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

Cov →

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

eigen values

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

Principal Component Analysis Solved Example 1

Step 4: Computation of the eigenvectors

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda I) U$$
$$= \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$
$$= \begin{bmatrix} (14 - \lambda)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda)u_2 \end{bmatrix}$$

We consider the first equation

$$\begin{aligned} (14 - \lambda)u_1 - 11u_2 &= 0 \\ -11u_1 + (23 - \lambda)u_2 &= 0 \end{aligned}$$

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U is the eigen vector
I is the identity matrix
S is the covariance

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

mean

$$\begin{aligned} \bar{X}_1 &= 8 \\ \bar{X}_2 &= 8.5 \end{aligned}$$

Cov →

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\begin{aligned} \lambda_1 &= 30.3849 \\ \lambda_2 &= 6.6151 \end{aligned}$$

Principal Component Analysis Solved Example 1

$$\begin{aligned} (14 - \lambda)u_1 - 11u_2 &= 0 \\ -11u_1 + (23 - \lambda)u_2 &= 0 \end{aligned}$$

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda} = t$$

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\frac{u_1}{11} = \frac{u_2}{14 - \lambda} = t$$

$$u_1 = 11t, \quad u_2 = (14 - \lambda)t$$

$$U_1 = \begin{bmatrix} 11 \\ 14 - \lambda \end{bmatrix}$$

Assume
 $t = 1$

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

Principal Component Analysis Solved Example 1

Step 4: Computation of the eigenvectors

- To find the first principal components, we need only compute the eigenvector corresponding to the largest eigenvalue.
- In the present example, the largest eigenvalue is λ_1 .
- So, we compute the eigenvector corresponding to λ_1 .

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\bar{X}_1 = 8$$
$$\bar{X}_2 = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$
$$\lambda_2 = 6.6151$$

We need only one principal component then we use λ_1 , if you need another principal component then you should use λ_2

Principal Component Analysis Solved Example 1

$$U_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix} \Rightarrow U = \begin{bmatrix} 11 \\ 14 - \lambda \end{bmatrix}$$

- To find a unit eigenvector, we compute the length of U_1 which is given by,

$$\begin{aligned} \|U_1\| &= \sqrt{11^2 + (14 - \lambda_1)^2} \\ &= \sqrt{11^2 + (14 - 30.3849)^2} \\ &= 19.7348 \end{aligned}$$

Length of the Unit eigen vector U_1

$$\begin{aligned} e_1 &= \begin{bmatrix} 11/\|U_1\| \\ (14 - \lambda_1)/\|U_1\| \end{bmatrix} \\ &= \begin{bmatrix} 11/19.7348 \\ (14 - 30.3849)/19.7348 \end{bmatrix} \\ &= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix} \end{aligned}$$

~~$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$~~

Principal Component Analysis Solved Example 1

Step 5: Computation of first principal components

$$\begin{aligned} e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} &= [0.5574 \quad -0.8303] \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{21} - \bar{X}_2 \end{bmatrix} \\ &= 0.5574(X_{11} - \bar{X}_1) - 0.8303(X_{21} - \bar{X}_2) \\ &= 0.5574(4 - 8) - 0.8303(11 - 8.5) \\ &= -4.30535 \end{aligned}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

$$\bar{X}_1 = 8$$

$$\bar{X}_2 = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

Principal Component Analysis Solved Example 1

Step 5: Computation of first principal components

Feature	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14
First Principle Components	-4.3052	3.7361	5.6928	-5.1238

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

Principal Component Analysis Solved Example 1

